

# ENGINEERING COMPONENTS FOR A SODIUM COOLED FAST POWER REACTOR

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## 1. Introduction

In the same way as the Rapsodie studies in France were followed by the study of large fast reactors, in particular the prototype reactor Phenix,<sup>(1)</sup> so the testing and development of the Rapsodie components<sup>(2)</sup> has been followed by examination of the problems raised by the components of large reactors.

The designs envisaged for these components are in general, as will be seen later, greatly influenced by our earlier work, in particular the apparatus built for Rapsodie.

The present position of our studies will be outlined with descriptions of the experience acquired or the tests in progress or to be undertaken.

## 2. Pumps

Centrifugal, vertical pumps with a free level will be used for Phenix. This type of pump, which is essential for a large power reactor, is used for Rapsodie.<sup>(3, 4)</sup>

Phenix will be equipped with three primary pumps. The pressure/delivery characteristics of each pump, in normal operation and with the flow reversed (impeller blocked), have been fixed so that in the event of accidental stoppage of one pump the two others will guarantee the passage of 80% of the nominal flow in the core. The pumps would therefore not be compulsorily fitted with a non-return valve.

Studies of different configurations were carried out by two French companies‡ to define the pumps for Phenix and those for a reactor of 1000 mW(e). For Phenix the pumps were designed for a delivery of 4800 m<sup>3</sup>/h and various possible delivery heads from 60 to 150 m. Two systems are envisaged: either

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an impeller with a single intake, rotating at 1000 rev/min, resulting in equipment 1.60 m in diameter, or an impeller with two symmetrical intakes rotating at 1500 rev/min, the diameter of the pump being reduced to 1.20 m in

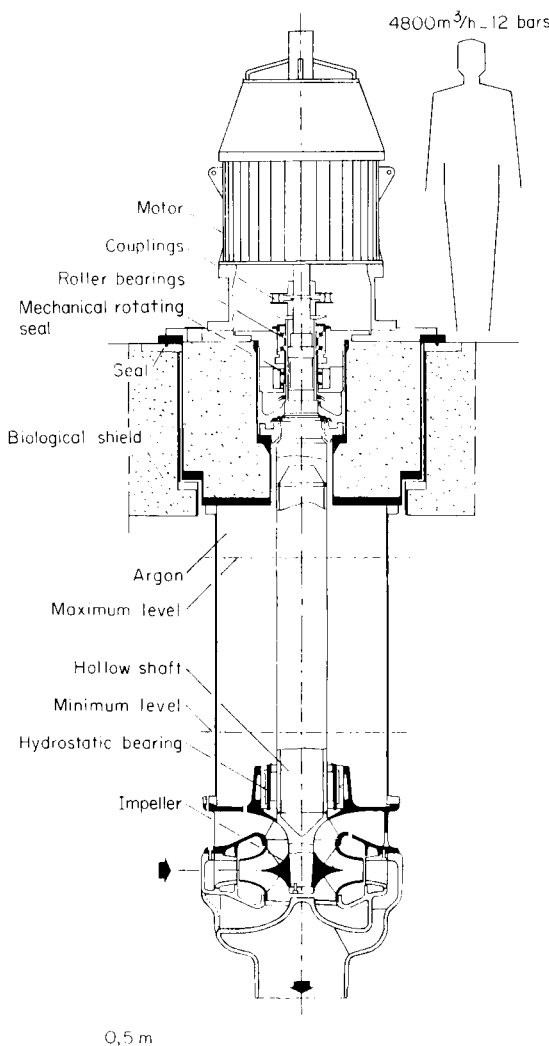


FIG. 1. Primary mechanical pump for Phenix.

that case. It is hoped to use the second system. In both cases the hydraulic efficiency would be of the order of 85 percent.

It has been estimated that for a reactor of 1000 MW(e) the pump diameter (9200 m³/h, delivery head 150 m) would be only 0.30 m or 0.40 m larger than the figures just given.

Figure 1 is a diagram of a pump for Phenix corresponding to the second system mentioned above, and Fig. 2 shows a comparison of the Phenix and Rapsodie pumps. The pump can be completely dismantled with the movable shield plug. The motor, the upper bearings, and the mechanical rotating seal

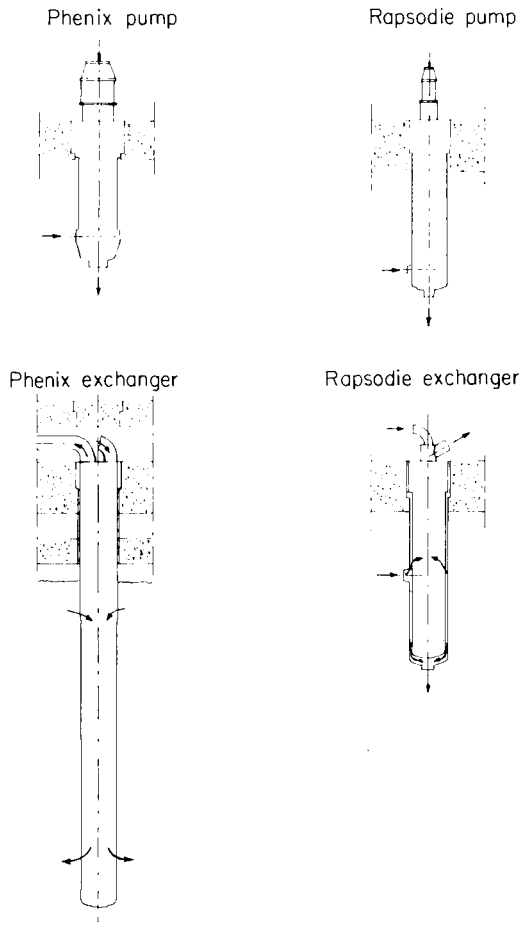


FIG. 2. Comparative dimensions of pumps and exchangers for Phenix and Rapsodie.

which ensures leak-tightness of the argon atmosphere on the rotating shaft, are accessible at ground level. The hollow shaft, about 3.5 m long, has a submerged hydrostatic bearing at the bottom, supplied with sodium by the delivery of the pump. The intake occurs freely from the sodium in which the pump is immersed; delivery is axial.

Our first mechanical sodium pump was built 10 years ago and more than fifteen pumps of all sizes have been constructed since then. A pump in one

of the Rapsodie test loops ( $50 \text{ m}^3/\text{h}$  approximately) operated for 16,000 hours without being dismantled, the only action taken being to change the upper leak-tight gasket. The pumps in the 10 MW loop (about  $500 \text{ m}^3/\text{h}$ ) have been operated for 13,000 hours. Finally a pump of about  $1000 \text{ m}^3/\text{h}$  has already operated for 3000 hours. The experimental investigations carried out elsewhere both on sodium bearings and on mechanical packings give us every confidence in the pumps designed for Phenix.

Provision has been made, however, for a prototype to be built soon, which will be tested to introduce modifications to the original design if required. This prototype could be fitted on the delivery side with a sliding and flexible device (bellows) designed to provide the junction between the removable pump and the fixed delivery connexion in the main vessel of an integral-circuit Phenix reactor.

Experiments have also been going on for several years on a pump for testing solidified sodium sealing on the shaft. The use of this type of pump could be considered for the secondary circuits of the reactor. It is still too early, however, to say whether it would be beneficial.

### 3. Intermediate Exchangers

Various types of exchangers have been considered for Phenix according to the configurations examined for the reactor and its circuits. Only the intermediate exchangers for the system with integral primary circuits will be described here. They are made up of bundles of vertical tubes, which can be dismantled, between two tube plates and immersed in the sodium of the main vessel (Fig. 3). There are six of these exchangers.

The secondary sodium circulates in the tube from bottom to top, entry and exit of the fluid being through pipes which enter the unit from above through a biological shield plug, integral with the bundle. The outlet pipe removes the secondary sodium from an upper header over the tube bundle. The inlet pipe drops along the axis of the equipment to an inlet header consisting of a chamber with a dished end situated at the bottom of the bundle.

The primary sodium passes round the tubes from top to bottom. The flow is guided by a sleeve surrounding the bundle and penetrated at the top and bottom by inlet and outlet orifices. The sleeve is integral with the bundle but can be moved independently, which allows it to be supported, by way of a circular ring, on the horizontal collar of the primary vessel of the reactor. This arrangement provides adequate leak-tightness to ensure circulation of the sodium inside the equipment. One alternative considered entails using a sleeve rigidly attached to the bundle and sliding vertically in a ring which can move horizontally.

This description applies in all respects to the Rapsodie equipment<sup>(3)</sup>

(Fig. 2), except that the outer shell is eliminated. In the design, performance, and construction of these exchangers, great benefit will therefore be derived

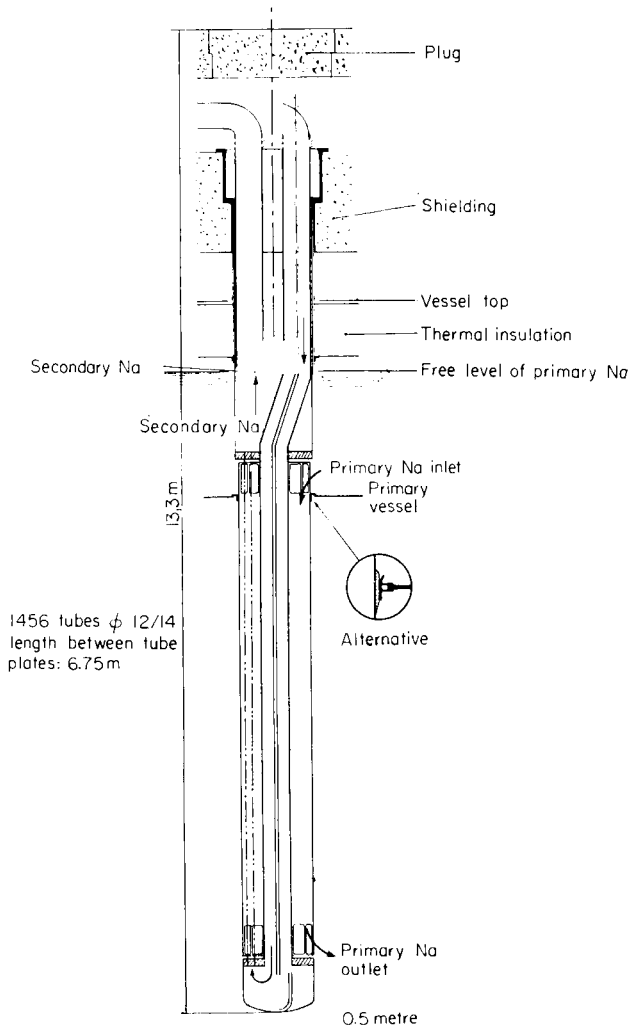


FIG. 3. Phenix intermediate exchanger.

from the experience acquired on the Rapsodie prototype exchangers (10 MW units), testing and operation of which have been in progress since 1962, and on the exchangers in the reactor.

The only differences in design would be the following:

- (a) In the Rapsodie exchangers fitted on the circuits outside the reactor, the position of the primary sodium level, which represents the pressure drop between the vessel and the exchanger, varies as a function of the rate of flow. As a result, a length of about 1 m of tube bundle is not used in normal operation. The integral circuit version for Phenix makes it possible to have a practically fixed level in the exchangers and therefore to use the total length of the tubes permanently.
- (b) It is hoped to eliminate in Phenix the individual expansion bends which are found on the tubes of the Rapsodie exchangers.

Thermal tests will probably be carried out to verify the stability of the tubes without expansion bends. Hydraulic tests will also be carried out to determine the homogeneity of the flow, definition of the supply orifices, pressure drop, and the system of holding the tubes to avoid any vibration.

Extrapolation of the exchangers to a 1000 MW(e) integral circuit reactor has also been examined. There would be eight units with about 4000 tubes each. The basic system would remain the same.

#### 4. Steam Generators

##### Studies for Phenix

Rapsodie, of course, is not fitted with steam generators, at least in its present design. Phenix, on the other hand, will be a prototype power producer, 250 MW(e) is the value taken as a basis for the studies.

The thermal parameters of the reactor, defined by a preliminary study,<sup>(1)</sup> have led to the following characteristics:

|                                |           |
|--------------------------------|-----------|
| Steam at admission to turbine: |           |
| Pressure (kg/cm <sup>2</sup> ) | 166       |
| Temperature (°C)               | about 540 |
| Water temperature (°C)         | about 280 |
| Reheat:                        |           |
| Pressure (kg/cm <sup>2</sup> ) | 34.2      |
| Temperature (°C)               | about 540 |
| Secondary sodium:              |           |
| Inlet temperature (°C)         | about 555 |
| Outlet temperature (°C)        | about 350 |

For the Phenix steam generators the systematic trend was towards once-through units with single walls.

Various configurations\* have been studied, calculated, and designed, which can be summarized as follows:

- (a) Concentric hairpin tubes. This is the unit nearest to the 5 MW prototype currently being tested at Grand Quevilly, except that the double

\* Contracts with Stein & Roubaix and G.A.A.A.

wall has been eliminated. The water and steam circulate in a tube surrounded by an annular space containing the sodium in a counter-current flow.

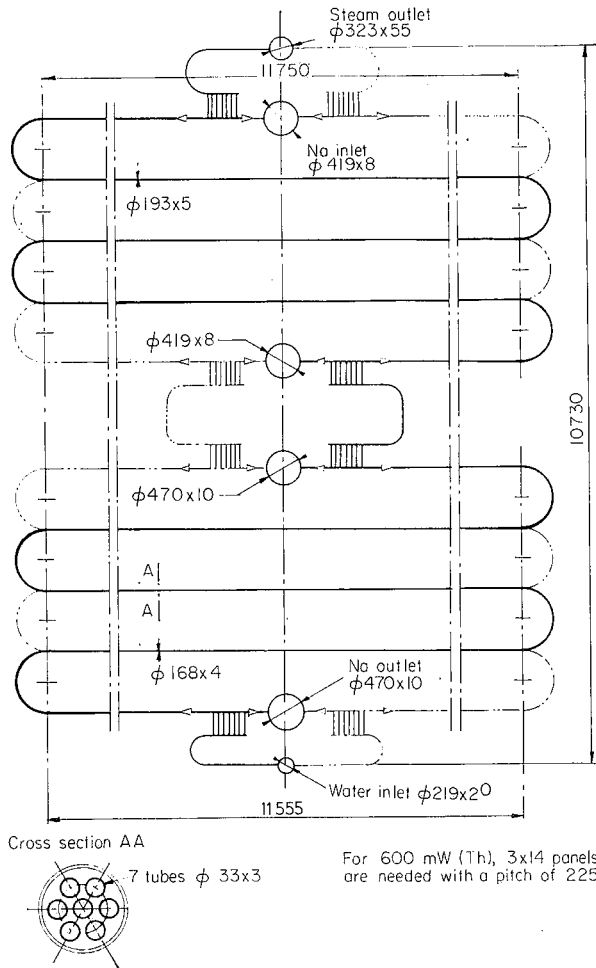


FIG. 4. Steam generator seven-tube hairpin configuration.

- (b) Bundle of tubes of limited number, housed in a cylindrical shell with a low cross-section, the whole shaped like a hairpin as in the previous configuration. This is an arrangement derived from the first one and characterized by the fact that all the tubes leave the cylinder individually; the unit therefore has no tube plate. Units with seven and twenty-one tubes have been specifically designed (Fig. 4).
- (c) Exchangers with straight tubes and tube plates. This version is the one

nearest in principle to the conventional heat exchangers. Several versions were considered, in particular the one developed by Atomics International.

- (d) Flat panels in a large diameter cylinder. The water and steam circulate in hairpin tubes forming parallel vertical panels. The sodium circulates

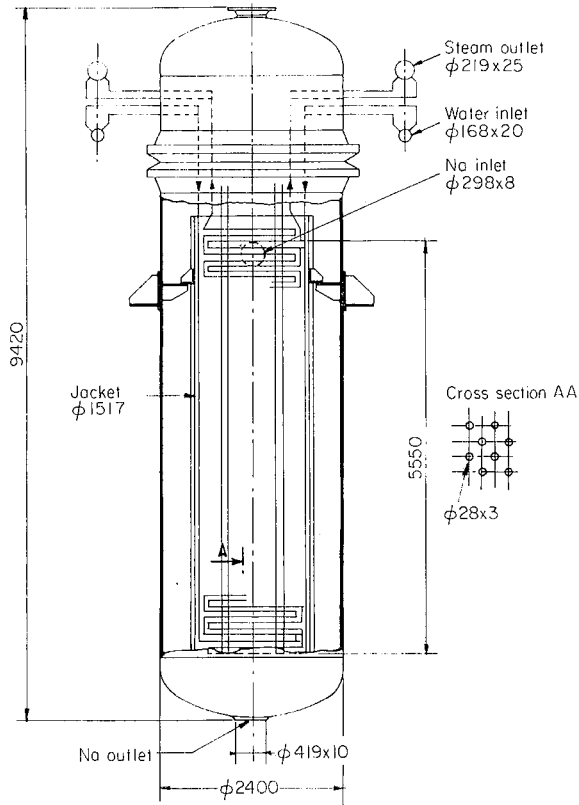


Fig. 5. Steam generator configuration with flat panels in a cylinder.

in the outer shell in a transverse flow (Fig. 5). The design of this configuration showed an opposite trend to the previous ones; it was intended to achieve a very limited number of units in order to limit the size of the whole and the water and sodium connexions.

The respective merits of the various configurations considered were compared from the following aspects:

- (i) accidental contact between sodium and water; risk of leakage, resist-



ance of the unit to accidental over-pressure, availability of the station after a damaged unit is isolated, ease of identification of a defective element, replacement of a damaged module by a new one;

- (ii) problems of operating behaviour; differential expansion, thermal shocks, supporting and vibration of tubes, draining, stability of flow, etc.;
- (iii) layout and connexions;
- (iv) extrapolation to a higher power (1000 MW(e));
- (v) price.

Comparative examination of the different versions studied is still in progress and it is too early to give a final verdict. It is also not expected that one conclusion will be drawn from this, but that at least two alternatives will be retained, prototypes of which will be built for testing.

Bearing in mind these reservations, it can, nevertheless, be said that a bundle system with a small number of tubes in a hairpin shell (Fig. 4) seems a useful one, at least at the present state of progress. This type of unit stands up favourably to the technical criteria mentioned above. Moreover, of the types examined, it is the one with the lowest relative price.

The materials at present considered are a 2.25 Cr steel for the evaporator and a 321 austenitic steel for the superheater and resuperheater.

### Testing of Prototypes

The essential feature of the French research programme on steam generators heated by circulating liquid metal is the 5 MW rig situated at Grand Quevilly near Rouen.<sup>(5)</sup> This rig, which came into operation in June 1964, reached its rated conditions in December 1964. It comprises essentially a primary loop with a reheater fed with no. 2 heavy fuel oil, enabling the sodium to be heated to a temperature of 600–625°C, a secondary NaK loop with a hot leg at a temperature of 585–600°C and a steam–water loop, the main equipment in which is an experimental steam generator.

The generator at present in service is a double walled unit designed in 1958 to produce steam under good safety conditions, the steam characteristics being considered fairly high at that time (545°C—129 kg/cm<sup>2</sup>). The unit consists of coaxial hairpin tubes<sup>(5)</sup> and was made of 321 austenitic steel both for the wall in contact with the water or steam and those in contact with the NaK alloy (56% K). The use of austenitic steel in contact with the water was possible because of the quality of water treatment: 25% of the flow of water through the steam generator is continuously retreated in an ion exchange column (mixed beds). In addition, injections of morpholine maintain a pH of more than 9 and periodic additions of hydrazine ensure a very low oxygen content. Currently, water with a resistivity of more than

$10^6 \Omega/\text{cm}$  is obtained, with a silica content less than  $30 \mu\text{g/l.}$  and chloride content less than  $50 \mu\text{g/l.}$

The leak-tightness of the generator after more than 2500 hours' service is excellent. Measurements of the hydrogen content of the blanket argon taken periodically by means of a chromatograph give values generally near to 50 vpm (volumes per million) for the argon over the intermediate NaK and 30 vpm for the argon in the secondary NaK expansion tank.

At the nominal steam pressure and flow conditions the temperature difference between the secondary NaK and the superheated steam is  $38^\circ\text{C}$  instead of the calculated  $40^\circ\text{C}$ , in spite of unequal distribution of the steam flow between the six tubes operating in parallel. These inequalities of flow, possibly connected with the relatively low pressure drop (around  $3 \text{ kg/cm}^2$ ) in the unit, have resulted in a control valve and flowmeter being added to the inlet of each water tube. By means of these valves it has been possible to reduce to less than  $2^\circ\text{C}$  the steam differences between the various tubes which, for certain conditions, attained  $50^\circ\text{C}$ . When the control system is operating on reactor priority the stability of the steam temperature at the outlet from the header is better than  $1^\circ\text{C}$ .

Operation of the steam generator raises no problems. The speed of start-up of the rig and of its rise to rated conditions is at present only limited by the achievement of a suitable silica content in the water, which requires about 48 hours (after each shutdown period of any length the silica content in the water circuit tends to rise above  $100 \mu\text{g/l.}$ ). Start-up takes place with the tubes full of water.

During the summer of 1966 the rig will be equipped with a pump delivering water to the generator at a maximum pressure of  $255 \text{ kg/cm}^2$ .

Following the tests on the first prototype steam generator (designed in 1958), it is intended to carry out tests from the end of 1967 on prototypes of the designs selected for the Phenix reactor. Apart from the study of the thermal performance, these tests will allow a check to be made on the pressure drops of the fluids, the operating stability, and the correct behaviour of the steels used.

### **Studies on the Consequences of Contact between Sodium and Water**

The design of the jackets and the reliability of sodium steam generators depend greatly, of course, on a correct knowledge of the effects of the sodium-water reaction in the event of an accidental leak. To study this problem a test rig has been operated at Cadarache for about a year.<sup>(6)</sup> It comprises a NaK (or Na) circuit and a steam-water circuit; the essential feature is a small 50 kW single-tube steam generator capable of producing superheated steam at  $550^\circ\text{C}$  at a pressure of  $128 \text{ kg/cm}^2$ . The unit is made up of two coaxial tubes, water and steam circulating in the inside tube, NaK in the outside space. An electric arc device enables the unit to be pierced at several points, a hole

TABLE 1. STEAM GENERATOR SAFETY EXPERIMENTS

*Characteristics of the test circuit*

Maximum power: 50 kW. Metal: 304 and 316 austenitic steel. Volume of liquid metal: 30 litres. Volume of argon in expansion tank: 11 litres. Diameter of main piping: 20 mm. Liquid metal: NaK (56%) (except test  $I_{1D}$ ; Na). Isolating equipment: pneumatic valves.

*Steam generator*

Type: once-through hairpin, single-wall. Metal: 321 austenitic steel. Exchange length: 8.4 m. Steam tube diameter: 20/23 mm. Number of steam tubes: 1. External diameter of steam tube: 30/34 mm.

| No. of test:                                                                   | $I_{1B}$          | $I_{1C}$   | $I_{2A}$               | $I_{3A}$               | $I_{4A}$    | $I_{1D}$   |                    |
|--------------------------------------------------------------------------------|-------------------|------------|------------------------|------------------------|-------------|------------|--------------------|
| Steam pressure (bars)                                                          | 127               | 120        | 120                    | 122                    | 127         | 125        | Initial conditions |
| Water inlet temperature (°C)                                                   | 205               | 200        | 185                    | 200                    | 185         | 206        |                    |
| Steam outlet temperature (°C)                                                  | 510               | 500        | 525                    | 485                    | 500         | 530        |                    |
| Liquid inlet temperature (°C)                                                  | 550               | 555        | 565                    | 550                    | 555         | 581        |                    |
| Liquid metal outlet temperature (°C)                                           | 250               | 250        | 265                    | 270                    | 250         | 258        |                    |
| Argon pressure (kg/cm <sup>2</sup> )                                           | 0.04              | 0.04       | 0.05                   | 0.05                   | 0.1         | 0.06       | Leak               |
| Liquid metal flow rate (m <sup>3</sup> /h)                                     | 0.45              | 0          | 0.396                  | 0.36                   | 0.3         | 0.3        |                    |
| Initial temperature (liquid metal at leak) (°C)                                | 320               | 260        | 370                    | 340                    | 480         | 320        |                    |
| Orifice cross-section (mm <sup>2</sup> )                                       | 2.4               | 0.9        | 2.3                    | 2.6                    | 2.5         | 5.1        |                    |
| Position of orifice                                                            | Economizer        | Economizer | Evaporator (1st point) | Evaporator (2nd point) | Superheater | Economizer |                    |
| Maximum leakage rate (kg/sec)                                                  | 0.188             | 0.11       | 0.15                   | 0.14                   | 0.037       | 0.4        | Results            |
| Duration of test (sec)                                                         | 7.4               | 2.9        | 2.8                    | 2.8                    | 2.9         | 3          |                    |
| Maximum temperature at leak (°C)                                               | 810               | 900        | 1090                   | 1030                   | >1200       | 1200       |                    |
| Wall temperature rise (°C)                                                     | 0                 | 0          | 0                      | 0                      | 25          | 10         |                    |
| Maximum temperature at discharge pipe (°C)                                     | 870               | 460        | 790                    | 900                    | 1050        | 710        |                    |
| Maximum pressure at leak $P$ (kg/cm <sup>2</sup> )                             | 90                | 20         | 40                     | 60                     | 51          | —          | Safety             |
| Maximum pressure of liquid metal circuit: $P_1$ or $P_2$ (kg/cm <sup>2</sup> ) | 80                | 10         | 25                     | 54                     | 12          | —          |                    |
| Maximum argon pressure in expansion tank (kg/cm <sup>2</sup> )                 | 1.1               | 2.2        | 1.4                    | 1.8                    | 0.4         | 2.7        |                    |
| Bursting disks M1 and M2 burst at                                              | 170 msec          | 270 msec   | 180 msec               | 160 msec               | 250 msec    | 120 msec   |                    |
| Automatic isolation of generator                                               | Manual at 7.4 sec | At 2.9 sec | At 2.8 sec             | At 2.8 sec             | At 2.9 sec  | At 3 sec   |                    |

2–3 mm inside diameter being produced through the wall of the inner tube containing the water and steam. The pressure due to the sodium–water reaction causes the unit to be isolated on the sodium side and the water side by pneumatic valves (closing time 50 msec). It also causes the rupture of the two bursting disks (50 mm in diameter and calibrated at  $5 \text{ kg/cm}^2$ ) with which it is fitted. The reaction products are brought to the outside via a discharge pipe.

The controls and measurements are effected from a cabin situated about 30 m from the test area.

Measurement of rapidly varying pressures is achieved by means of piezo-electric pickups with a range extending from 0 to  $255 \text{ kg/cm}^2$ . We thus know the pressure  $P$  in the vicinity of the point of penetration and the pressures  $P_1$  and  $P_2$  at the two ends of the generator. The temperatures during the test are measured by three thermocouples sheathed in stainless steel (0.25 mm in diameter) and immersed directly in the liquid metal. The measurements are recorded on two oscilloscopes fitted with photographic equipment.

In 1965, eight NaK–H<sub>2</sub>O reaction tests were carried out and one Na–H<sub>2</sub>O test, the results of which are given in Table 1 (for two tests the measurement recording system was not working).

From the pressure  $P$  measured near the point of penetration we attempted to discover by calculations the development of the pressures  $P_1$  and  $P_2$  measured from the discharge pipes fitted with a bursting disk. Figure 6 shows the results obtained. The oscillating character of the pressure  $P_1$  and  $P_2$  measured is undoubtedly due to the reflections at individual points of the pressure waves emitted near the point of penetration.

The results of the temperature measurements made in the reaction zone were found by a simplified thermodynamic calculation with differences of about  $70^\circ\text{C}$ .

It can be concluded from this first series of experiments that:

- (a) the temperature of the reaction products is an increasing function of the steam quality; the maximum temperature measured for a leak of superheated steam is about  $1300^\circ\text{C}$  (test I<sub>4A</sub>) (Fig. 6);
- (b) the maximum pressure is attained in the case of a leak of water under slight saturation ( $90 \text{ kg/cm}^2$ ) in test I<sub>1B</sub>;
- (c) the increase in the wall temperature of the outer tube near the leak is negligible ( $25^\circ\text{C}$  at the maximum);
- (d) in the case where 321 austenitic steel is used the reaction products do not corrode the wall during the very brief period for which they remain in the apparatus;
- (e) the use of bursting disks and rapid-closing isolating valves makes it possible to limit the effects of a leak to quite acceptable consequences.

The present experimental circuit will be modified to permit study of extremely small leaks lasting a long time (corrosion effects).

A new circuit of 100 kW is being built in which much larger leaks will be produced (complete fracture of a tube with an internal diameter of 15 mm)

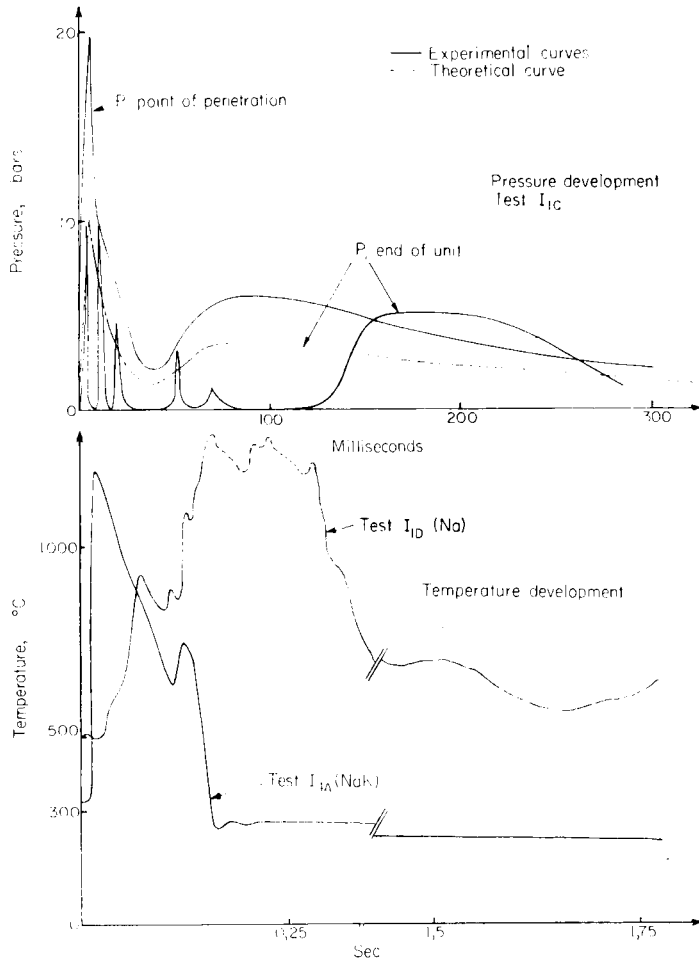


FIG. 6. Study of the sodium-water reactor.

in order to observe the effect of the reaction on the adjacent tubes in a bundle. The pressure of the water circulating by natural convection in the tubes will be of the order of 170 bars and the temperature will be near 350°C.

Finally, tests will be carried out later on elements of the steam generators

chosen for Phenix. As the information obtained from the present experiments has shown that the most difficult parameter to find by calculation is the rise in pressure near the leak, it is intended in future to turn to full size tests in respect of the cross-section of the equipment but on relatively short lengths.

### 5. Fuel Handling

In another paper<sup>(1)</sup> reference has been made to the studies under way for defining the Phenix fuel handling system. The information obtained from tests on the Rapsodie handling mechanisms will be briefly indicated here.

Rapsodie uses a system of two rotating plugs, a large plug centred on the axis of the reactor and a small offset plug inside the first one.<sup>(1)</sup> A handling channel vertically through the small plug can be brought over any fuel assembly by appropriate rotation of the two plugs. A charge-discharge machine is placed over the handling channel to withdraw a spent assembly and introduce a new assembly. The assemblies are brought out directly without internal storage in the reactor vessel.

A considerable number of handling operations have been carried out for more than 2 years on the full sized vessel mockup, under conditions as similar as possible to those in the reactor.

The following general conclusions, which will be borne in mind for Phenix, have been reached:

- (i) it is preferable to use many simple mechanisms each with a limited role rather than a multi-purpose device;
- (ii) it is desirable for the handling channel to occupy a fixed position during charging or discharging. The machines are then brought to fixed points, which eliminates the difficult adjustments before connexion to the channel;
- (iii) the problem of cooling an assembly during discharge is a delicate one; it is necessary to avoid direct cooling of the assembly. For Phenix it is intended to discharge the assemblies into a pot. In this way the entrainment of sodium is reduced as a result of the simple shape of the wall; the sodium in the pot provides substantial thermal inertia; finally, heat exchange is easier than with a bare assembly;
- (iv) the clearances in the mechanisms should be comfortable; the insertion or extraction of the assemblies does not need to be done with great accuracy; a tolerance of  $\pm 10$  mm is planned for Phenix;
- (v) any mechanism operating in the reactor vessel must be capable of dismantling when necessary. If it is not irradiated, however (internal neutron shielding), it does not need to be systematically dismantled after each series of charge-discharge operations;

- (vi) it does not seem necessary to rest on the tops of the assemblies adjacent to the one being discharged especially in the case of heavy assemblies (200–300 kg) such as those for Phenix. A tube can be kept for guiding the vertical movement of the assembly, but without resting on the adjacent assemblies;
- (vii) only one case of jamming of the rotating plugs was observed on the Rapsodie mockup after a shutdown of 4 months. This was eliminated by heating the plugs, which proves that it was due to sodium deposits. It is therefore necessary to adhere to a plug design which minimizes such deposits;
- (viii) the liquefiable metal seals (lead–tin alloy) showed surface oxidation (formation of black powder), but leak-tightness was maintained. No increase in the melting point was observed;
- (ix) the lip seals, of the EFFBR type, have given satisfactory results but are fragile and difficult to fit. Inflatable rubber O-ring seals are preferred to these.

## 6. Other Components

### Cold Traps

For the secondary circuits of Phenix a similar arrangement to that of the secondary traps of Rapsodie<sup>(7)</sup> will probably be used, since it has given every satisfaction on the Rapsodie prototype circuits, even after considerable contamination of the circuits; this situation may arise in the event of a substantial leak in a steam generator. The trap has an internal economizer.

The primary cold traps are not designed to be capable of dismantling; it is intended to place them in concrete cells in which they will be discarded when they are full of oxide. This design will make it possible to avoid the expensive mobile shielding of the Rapsodie traps, and the complications involved in dismantling them in anticipation of possible high activity. It also allows the volume of the traps to be increased without too much expense and consequently enables part of the economizer to be incorporated in it, which greatly facilitates operation.

At present the possibility is being examined of constructing primary traps arranged horizontally, without an internal lining. A greater oxide capacity is expected from this as a result of substantial internal natural convection conveying the oxide to a region where there would be no risk of it blocking the main flow. Tests will be carried out on prototype equipment.

### Plugging Indicators

Completely automatic plugging indicators have been developed for several years, giving continuously the value of the sodium plugging temperatures;

the cooling is controlled by the sodium flow to keep the temperature such that the equipment is constantly in the partial plugging condition.<sup>(7, 8)</sup>

### Flowmeters

Electromagnetic flowmeters either using a permanent magnet or an electromagnet are practical and reliable devices. Those fitted on the main circuits of Rapsodie are 200 mm in diameter.

For very large diameters (of the order of 600 mm or above) a system of flow measurement has been devised and tested using a small electromagnetic flowmeter fitted on a by-pass between the upstream and downstream end of a pressure drop in the main piping. Several alternatives have been tested: the aim is to find the arrangement giving the best linearity of by-pass flow with respect to total flow.

As regards the primary flowmeters of an integral circuit reactor, it is difficult to use permanent magnets as a result of the loss of magnetization at high temperature. An electromagnetic flowmeter with self-excitation has been studied and designed. A prototype will be built for testing.

### Level Measurements

For many years various types of level-measurement devices have been developing, either with discontinuous measurement (indicating a given level) or continuous (permanent measurement of the level). The discontinuous indicators<sup>(3)</sup> used already on Rapsodie are now fully developed.

For continuous measurement the following have been examined: (a) resistance devices;<sup>(3)</sup> (b) coupled inductance devices.

For the resistance measuring instruments the aim has been to perfect the compensating resistance circuit, the purpose of which is to make the signal insensitive to temperature variations. Attempts were also made to eliminate the perturbations due to condensation of sodium on the walls of the measuring tubes; for this, periodic washing was envisaged by raising the level (maintenance), and on the other hand internal heating was envisaged to eliminate the risk of condensation. Prototypes with internal heating are to be tested.

For the coupled inductance devices a system was developed for compensating temperature effects, comprising in particular a feedback or degenerative amplifier according to the materials used.

### Valves

The valves which we have used for many years on the experimental circuits are closed by means of a simple clack and sealed by means of a bellows on the spindle. These valves are satisfactory for limited use but the cut-off at



closure (upstream–downstream) is rarely perfect, especially on large diameter piping, and the leak-tight bellows are fragile components.

On Rapsodie, therefore, valves have been eliminated in the primary circuits proper and their use has been limited to the auxiliary circuits, the piping diameters in which do not exceed 100 mm.

For the reactors of the future, sodium valve technology is being developed mainly for isolation of the steam generators. The investigations have revealed the value of double disk valves and the system of sealing at the spindle by a solidified seal (system widely used by Atomics International). Six double disk, solidified seal, prototype valves have been tested, for passage diameters of from 30 to 100 mm. A model of the butterfly valve type has also been tested: the simplicity of design and the smooth control make this type very suitable for automatic control.

Finally, two automatic rapid closing prototypes are to be tested on the rigs for studying the sodium–water reaction.

## 7. Conclusion

We have described the state of our engineering knowledge on the main components of large power fast reactors, in particular the systems which will be used for Phenix as envisaged at present. The studies on these components were carried out under the Euratom–CEA Contract of Association.

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